MILITARY SPECIFICATION SHEET

# ELECTRON TUBE, VIDICON

## TYPE 7038

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: One inch, for film or live pickup with color or black-and-white TV cameras

PIN CONNECTIONS AND DIMENSIONS: See figure 1

ABSOLUTE RATINGS: See note 1

| Parameter: | Ef | DIsj | isjT | Ec3 | Ec2 | Ec1 |
|---|---|---|---|---|---|---|
| Unit: | V | µAdc | µa | Vdc | Vdc | Vdc |
| Maximum: | 6.9 | 0.25 | 0.55 | 750 | 750 | 0, -300 |
| Minimum: | 5.7 | --- | --- | --- | --- | --- |

TEST CONDITIONS: 6.3 (Note 2) (Note 3) (Note 4) 300 (Note 5)

ABSOLUTE RATINGS: See note 1

| Parameter: | Ehk | FI | T (faceplate) | tk | I (focus coil) |
|---|---|---|---|---|---|
| Unit: | Vdc | fc | °C | sec | mAdc |
| Maximum: | 10, -125 | 1,000 | 71 | --- | --- |
| Minimum: | --- | --- | 0 | --- | --- |

TEST CONDITIONS: --- (Note 3) 29 to 35 300 33

TEST PATTERN: EIA resolution chart, 1956, or equivalent. (See note 6.)

SCAN SIZE: See note 7

TEST COMPONENTS: See note 8

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GENERAL:

Qualification - Required

Ⓐ denotes changes



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| METHOD | REQUIREMENT OR TEST | NOTES | CONDITIONS | SYMBOL | LIMITS | | UNIT |
|---|---|---|---|---|---|---|---|
| | | | | | MIN | MAX | |
| | Qualification inspection | | | | | | |
| 1331 | Direct-interelectrode capacitance | - | Target to all; no voltages | C | 3.5 | 5.5 | pF |
| | Quality conformance inspection, part 1 | | | | | | |
| --- | Target quick age | 10,11, 26 | | --- | --- | --- | --- |
| --- | Alinement horizontal | 10,12, 13 | Esj/DIsj = 0.02 µAdc | --- | -36 | +36 | mAdc |
| --- | Alinement vertical | 10,12, 13 | Esj/DIsj = 0.02 µAdc | --- | -36 | +36 | mAdc |
| --- | Cutoff | 10,13, 14 | Esj/DIsj = 0.02 µAdc; FI/IsjT = 0.22 µAdc | Ec1 | -50 | -95 | Vdc |
| --- | Center resolution (1) | 10,13, 15,16 | Esj/DIsj = 0.02 µAdc; EI/IsjT = 0.22 µAdc | --- | 650 | --- | Lines |
| --- | Corner resolution | 10,13, 16,17 | Esj/DIsj = 0.02 µAdc; FI/IsjT = 0.22 µAdc | --- | 350 | --- | Lines |
| --- | Spurious signal | 10,12, 13,16, 18,26 | Esj/DIsj = 0.02 µAdc; FI/IsjT = 0.22 µAdc | --- | --- | --- | --- |
| (A) --- | Target volts (1) | 10,13, 19 | Esj/DIsj = 0.02 µAdc; FI = 0 fc | Esj | 25 | --- | Vdc |
| --- | Sensitivity (1) | 10,13, 20 | FI = 1.0 fc; Esj/DIsj = 0.02 µAdc | Isj | 0.063 | --- | µAdc |
| --- | Lag | 10,13, 21 | Esj/DIsj = 0.02 µAdc; FI/IsjT = 0.22 µAdc | Lag | --- | 25 | % |
| --- | Maximum grid No. 2 current | 26 | Esj = open; Ec1 = 0; Ec3 = 250 Vdc | Ic2 | 1,200 | --- | µAdc |
| --- | Gas ratio | 22,26 | | GR | --- | 0.12 | nAdc/ µAdc |
| (A) --- | Maximum dark current uniformity | 10,13, 23 | Esj/DIsj = 0.1 µAdc; FI = 0 fc | ΔDIsj | --- | 23 | % |
| (A) --- | Center resolution (2) | 10,13, 15,16 | Esj/DIsj = 0.1 µAdc; FI/IsjT = 0.4 µAdc | --- | 500 | --- | Lines |
| (A) --- | After image | 10,13, 24 | Esj/DIsj = 0.1 µAdc; FI/IsjT = 0.4 µAdc | --- | --- | 3 | Sec |
| (A) --- | Target volts (2) | 10,13, 19 | Esj/DIsj = 0.1 µAdc; FI = 0 fc | Esj | --- | 90 | V |
| (A) --- | Sensitivity (2) | 10,13, 20 | Esj/DIsj = 0.1 µAdc; FI = 0.1 fc | Isj | 0.023 | --- | µAdc |
| (A) --- | Raster burn | 10,13, 25,26 | Esj/DIsj = 0.1 µAdc; FI = 0 fc | --- | --- | --- | --- |
| | Quality conformance inspection, part 2 | | | | | | |
| 1301 | Heater current | - | | If | 560 | 640 | mA |
| 1105 | Permanence of marking | - | | --- | --- | --- | --- |



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| METHOD | REQUIREMENT OR TEST | NOTES | CONDITIONS | SYMBOL | LIMITS | | UNIT |
|---|---|---|---|---|---|---|---|
| | | | | | MIN | MAX | |
| | Quality conformance inspection, part 3 | | | | | | |
| --- | Life test | 9 | Group D; t = 500 hours (minimum) | --- | --- | --- | --- |
| --- | Life-test end points: | | | | | | |
| | Maximum grid No. 2 current | - | | Ic2 | 900 | --- | µAdc |

NOTES:

1. The symbols used in this specification, in addition to those in MIL-E-1, are defined as follows:

   Esj = Target voltage
   Isj = Signal-output current
   DIsj = Dark current
   IsjT = Target current (Sum of Isj and DIsj)
   isjT = Peak target current
   FI = Faceplate illumination
   T (faceplate) = Temperature of faceplate
   I (focus coil) = Focus coil current

2. With zero incident light on tube face, the target voltage (Esj) shall be adjusted to obtain the specified dark current (DIsj) (Esj/DIsj), as specified under test conditions.

3. The faceplate illumination (FI) shall be adjusted to obtain specified target current (IsjT) as specified under test conditions. This may be done by adjustment of the lens opening. IsjT is defined as the sum of the signal-output current component (Isj) (0.20 µA) and the DIsj.

4. Grids No. 3 and No. 4 are connected together internally. The voltage applied to these two elements shall be referred to as Ec3. Ec3 voltages apply with the use of the specified focusing coil current of 33 mAdc, producing a field strength of approximately 40 gauss. Ec3 and optical focus shall alternately be adjusted to obtain best focus (sharpest lines).

5. Adjust Ec1 until the beam current is just sufficient to discharge the photo-layer as observed by presence of detail in the highlights of the monitor.

6. The specified test pattern shall be placed on a transparent slide and shall be illuminated from the rear of the slide.

7. Standard EIA (RETMA) television scan shall be employed. The scan size shall be 1/2 x 3/8 inch (12.70 x 9.53 mm).

8. Test components.

   (a) Deflection yoke - Cleveland Electronics No. VY-111-3, or equivalent.
   (b) Focusing coil - Cleveland Electronics No. VF-115-5, or equivalent.
   (c) Alinement coil - Cleveland Electronics No. VA-118, or equivalent.
   (d) The above components shall be mounted as shown on figure 3.

9. Life-test conditions.

   (a) Adjustment for scan size is made by the use of a specially designed vidicon tube with a phosphor screen and with 300 Vdc applied to the target. Use RCA deflection yoke No. 208007, or equivalent, focusing coil No. 208005, or equivalent, and alinement coil No. 208006, or equivalent.
   (b) With the scan size adjusted to approximately 1/2 x 3/8 inch (12.70 x 9.53 mm) (do not allow scanned area to touch target ring), faceplate temperature of 45° ±5°C, adjust Ec1 to obtain 0.5 mAdc of grid No. 2 current. Cover left half of faceplate (looking into tube face with key pin up) with black tape, and adjust target voltage to obtain 0.20 µAdc (DIsj) with no light incident on the faceplate.
   (c) Adjust the incident illumination on the faceplate to obtain 0.40 µAdc target current (IsjT). (Light plus dark current.)
   (d) The target voltage shall be readjusted for specified dark current within 24 hours after the tube has reached the specified operating temperature. The incident illumination shall be readjusted when Esj is readjusted.

Ⓐ 10. The tube shall be tested in camera with associated monitors as shown in method 5426.



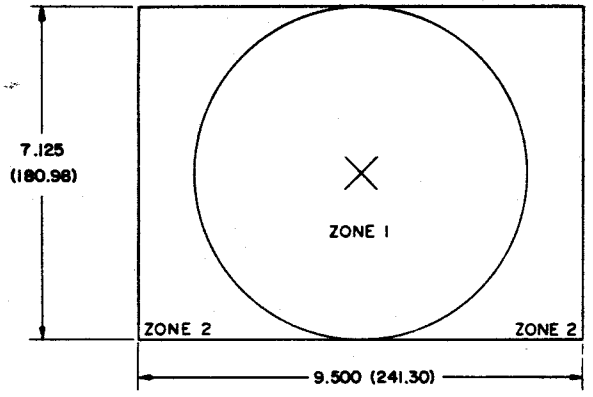
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NOTES: -Continued

11. Target quick age.

    (a) With no light on the tube, adjust horizontal, and vertical camera deflection, Ec3 and focus coil current to just overscan the target.
    (b) Set the target voltage at 300 volts and slightly defocus by adjusting Ec3. After 300 volts a lower target voltage is permissible, if burn can be removed.
    (c) Adjust Ec1 to obtain a complete white circular area on monitor with no distortion or dark areas present. The target current should read between 1 and 3 μAdc.
    (d) Age target for a minimum of 1 minute if target burns are encountered.

12. The following "back illuminated" test pattern shall be used when performing this test:




13. The alinement coil currents of each horizontal and vertical coil shall be adjusted until the center of the test pattern remains stationary as Ec3 is varied in and out of focus. The alinement coil currents shall be within the specified limits.

14. With test conditions as indicated, remove blanking voltage from grid No. 1 and adjust Ec1 until test pattern just disappears as observed on monitor.

15. With Ec3 adjusted for balanced horizontal and vertical resolution, the resolution in the center of the pattern shall exceed specified limit.

16. Cap lens and adjust black level for 20-percent setup. Uncap lens and adjust video gain control to see all steps of grey scale on resolution chart.

17. With Ec3 adjusted for balanced horizontal and vertical resolution, the resolution in the four corners of the pattern shall exceed the specified limit.

18. Spurious signal criteria.

    (a) To be considered a black spot, the contrast ratio must be greater than 2.0 to 1.0.
    (b) To be considered a white spot, the contrast ratio must be greater than 1.5 to 1.0.



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NOTES: -Continued

18. Spurious signal criteria. -Continued

(c) Spot size is measured by one of the three methods shown below.

| METHOD I<br>Measurement in inches using a monitor with 10-3/4 x 8-1/8 inch raster (273.05 x 206.38 mm) | METHOD II<br>Percent of raster height | METHOD III<br>Number of raster lines | NUMBER ALLOWED | |
|---|---|---|---|---|
| | | | Zone 1 | Zone 2 |
| Over 0.065 (1.65 mm) | Over 0.80 | Over 4 | None | None |
| 0.065 (1.65 mm) to but not including 0.050 (1.27 mm) | 0.80 to but not including 0.62 | 4 but not including 3 | None | 1 |
| 0.050 (1.27 mm) to but not including 0.015 (0.38 mm) | 0.62 to but not including 0.19 | 3 but not including 1 | 2 | 3 |
| 0.015 (0.38 mm) and under | 0.19 and under | 1 and under | * | * |

*Do not count spots of this size unless concentration causes a smudge appearance.

(d) Maximum concentration of spots shall be evaluated by one of the three methods shown below:

Method I: Minimum separation between any 2 spots greater than 0.015 inch (0.38 mm) is limited to 1/4 inch (6.35 mm).

Method II: Minimum separation between any 2 spots greater than 0.19 percent of raster height is limited to a distance equivalent to 3.1 percent of raster height.

Method III: Minimum separation between any 2 spots greater than 1 raster line is limited to 16 lines.

(e) Tubes are rejected for smudge, lines, streaks, mottled background, grainy background, or uneven background having contrast ratios greater than 1.5 to 1.0.

19. Recheck Esj after alinement is completed.

Ⓐ 20. Project specified faceplate illumination on vidicon target. The lamp current of the light source shall be adjusted for a color temperature of 2,856°K. Sensitivity is the target current under these conditions minus dark current.

21. Lag test.

(a) Equipment:

Ⓐ (1) Calibrated light source of 2,856°K color temperature.
(2) Suitable lens and apertures to obtain specified pattern and faceplate illumination.
(3) Motor-driven light chopper capable of interrupting the path of light four times per second. The light chopper shall provide light to the vidicon target for 1/5 second and interrupt for 1/20 second such that the vidicon target receives light for about 12 TV fields and receives no light for about 3 TV fields.
(4) An oscilloscope capable of monitoring the signal output including the dc component. (P7 screen is recommended.)

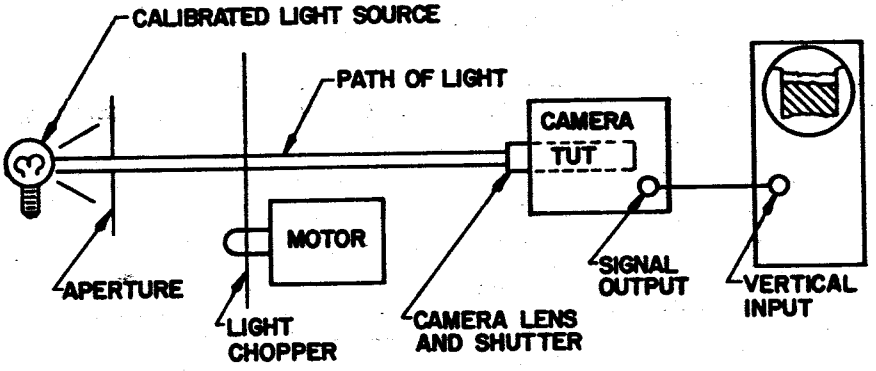


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NOTES: -Continued

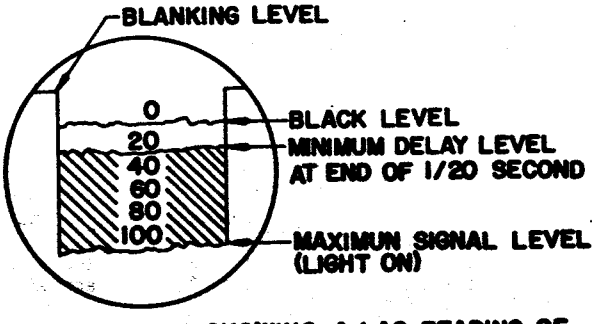
21. Lag test. -Continued

(b) Block diagram:




(c) Lag test procedure:

With conditions as indicated, and with test setup as shown on block diagram of paragraph (b), adjust oscilloscope sweep frequency to 60 Hz and adjust phasing control to obtain stationary pattern of signal output. Remove incident light on target and position horizontal trace on oscilloscope to coincide with "0" marking. Restore specified light level on target and adjust vertical gain and centering controls until pattern coincides with "100" marking. Turn on light-chopper motor and obtain pattern similar to that shown below:




TYPICAL OSCILLOSCOPE PATTERN SHOWING A LAG READING OF APPROXIMATELY 20 PERCENT.



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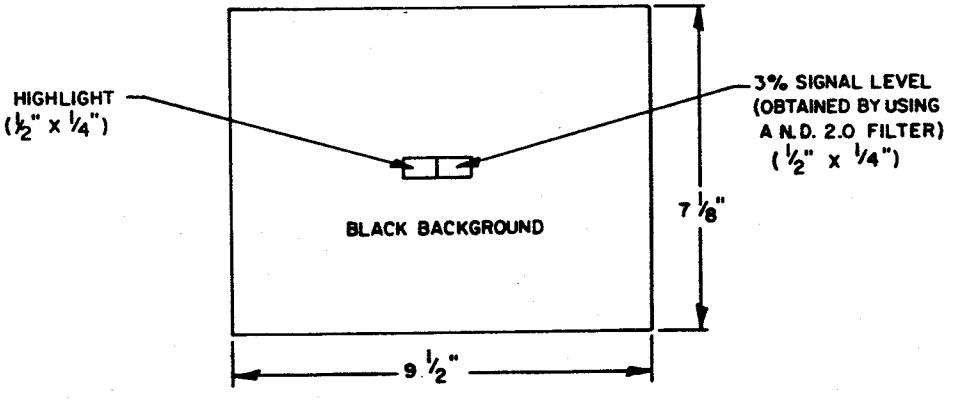
22. Gas ratio shall be determined as follows: With no light incident on the tube face, focus coil current "off", scanning "on", Esj open, Ec1 = 0 Vdc, the following tests shall be made:

| Ef | Ec3 | Test procedure | Units |
|---|---|---|---|
| 6.3 V | 250 Vdc | Record G3 current as PIC3. | µAdc |
| 6.3 V | -100 Vdc | Record G3 current as NIC3 when leakage becomes relatively stable, but not necessarily zero. | nAdc |
| 0 V | -100 Vdc | Turn filament off and record G3 current as ILC3 when G2 current drops to 0.5 mAdc. | nAdc |

Gas ratio (GR) = (NIC3-ILC3)/PIC3 nAdc/µAdc

23. The oscilloscope is adjusted looking at horizontal waveform, so that dark current signal above zero current (beam off) level is 100 percent on graduated scale. Total variation in dark current signal is measured as percentage of total signal. Ignore the few bright lines at top of picture and the few dark lines near bottom of picture.

24. Using the spurious signal test pattern as shown in note 12, set up tube to specified conditions. Replace above test pattern by the "After image test pattern" as shown below. The highlight brightness of the after image test pattern shall be the same as the highlight brightness of the spurious signal test pattern. Line select the waveform monitor trace through the center information. (Use IEEE response to reduce noise.) Adjust gain to make peak black-to-white signal level equal to 100 percent. Using the step gain switch, increase the signal amplitude to where a 3 percent signal level can be clearly seen. Expose the vidicon to the test pattern for an exact 30 seconds. Cap up the tube. Record the time (in seconds, using a stopwatch) required for the residual peak white signal to decay to a 3 percent level (above black) as after image.






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NOTES: -Continued

25. Raster burn test.

(a) With conditions as shown for 1/2 x 3/8-inch (12.70 x 9.53 mm) scan, completely overscan target and increase bias until complete target is discharged.
(b) Set black level at a value where no part of the dark current signal is clipped.
(c) Increase monitor bias until complete target is just visible.
(d) If a raster burn or image burn is present, repeat target quick age test.
(e) Repeat steps (a), (b), and (c).
(f) Reject if raster burn is lighter than surrounding area. Reject for any permanent image burn.

26. This test to be performed at the conclusion of the holding period.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

User activities:
Army - MU
Navy - AS, OS, MC, CG, SH
Air Force - 11

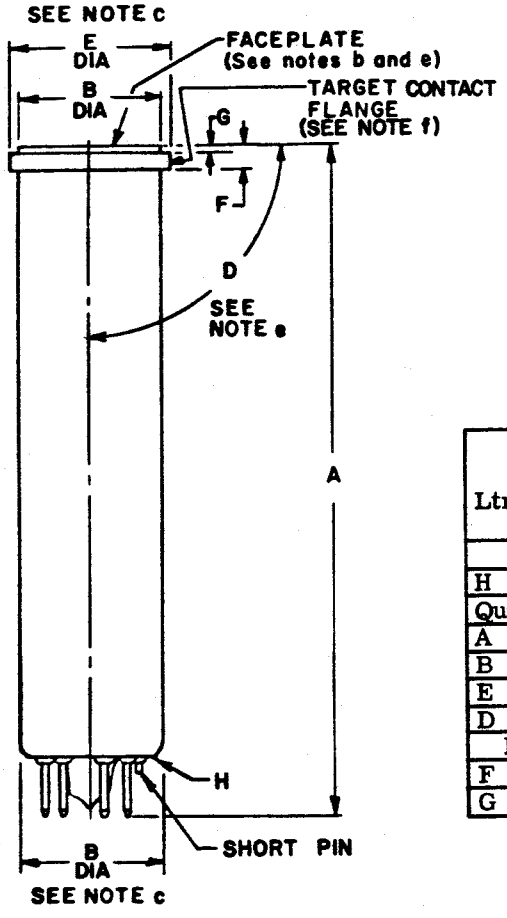
Preparing activity:
Navy - EC

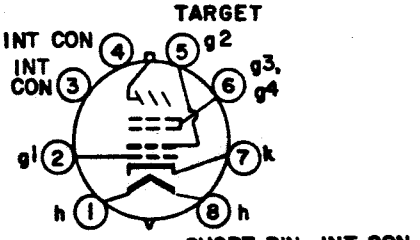
Agent:
DSA - ES

(Project 5960-2906)



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BASING DIAGRAM

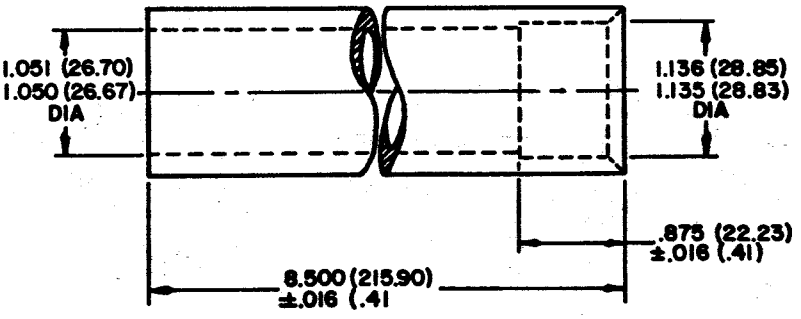
| Ltr | Dimensions in inches with metric equivalents (mm) in parentheses | |
|---|---|---|
| | Minimum | Maximum |
| Qualification inspection | | |
| H | Base: E8-11 (see note d) | |
| Quality conformance inspection, part 2 | | |
| A | 6.120 (155.45) | 6.380 (162.05) |
| B | .990 (25.15) | 1.045 (26.54) |
| E | 1.116 (28.35) | 1.132 (28.75) |
| D | 89.75° | 90.25° |
| Reference dimensions (see note a) | | |
| F | .175 (4.45) | |
| G | .050 (1.27) | |

NOTES:
a. These dimensions are for information only and are not required for inspection purposes.
b. Direction of incident light is into face end of tube.
c. Concentricity and camber of tube shall be determined by gage shown on figure 2.
d. For pin alinement, use gage GE8-2.
e. Faceplate tilt shall be gaged about entire circumference.
f. Target contact flange in form of metal ring, and with diameter as specified for dimension E, may be located along tube axis in any part of, or all of, space between dashed lines.

FIGURE 1. Outline drawing of electron tube type 7038.



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NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest . 01 mm) are given for general information only and are based upon 1 inch = 25. 4 mm.
3. Metric equivalents are in parentheses.

FIGURE 2. Gage, concentricity, and camber.

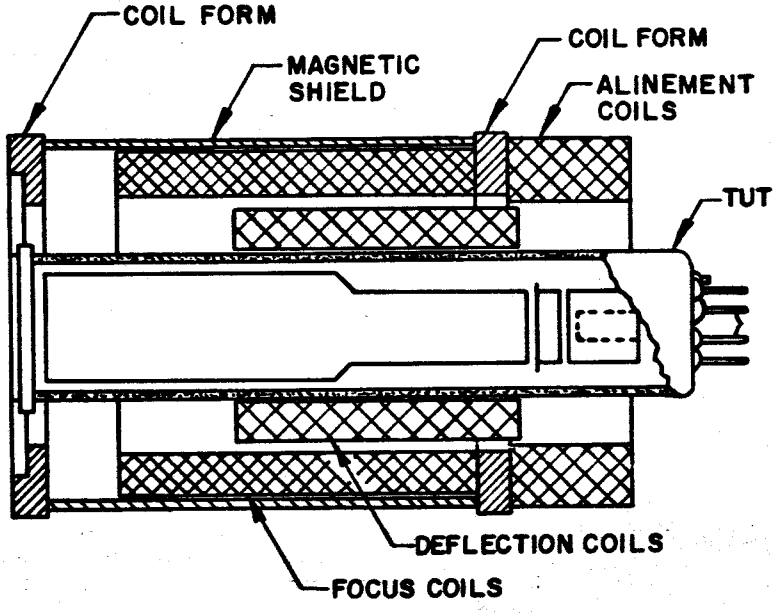



FIGURE 3. Test components.



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